

SRS RAMS

G-PRD S06000 REV.02

GBMS 2.0	COUNTRY
	ALL
DOCUMENT TYPE	LANGUAGE
PROCESS DESCRIPTION based on G-TMP A01007 rev.05	EN

Hitachi Rail – Internal

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REVISION HISTORY

Rev.	Date	Author	Approver	Authorizer	Revision Description
00	Mar 2022	O. Samuel Espin	R. Castagna	P. Marmo	First Issue
01	May 2023	O. Samuel Espin	R. Castagna	P. Marmo	Alignment with new GBMS Process Map. Updated interfaces with Service & Maintenance, S&T Asset Operation, S&T Manufacturing and S&T Requirements & Interface Management.
02	Apr 2026	O. Samuel Espin A. Antal B. Dogaru A. Orazzo	R. Castagna	O. Hocheid	(i) Migration to GBMS 2.0 (ii) Interfaces Updated (iii) Activities updated

DOCUMENTS TREE

L1. G-TRN S06000 RAMS Transition Rules

L1. G-PRD S06000 RAMS

L2. G-KPI S06001 RAMS KPIs

L2. G-PRC S06005 Manage Technical Safety Incidents

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1 SCOPE

1.1 Process Cluster

☐ Steering ☐ Support ☒ Execution ☐ Customer Facing

1.2 Purpose

The SRS RAMS process provides the framework of activities and related responsibilities to ensure the demonstration that a product or product component, a signalling system or subsystem, a turnkey system meets its specified Reliability, Availability, Maintainability, and Safety requirements, according to the applicable standards and contract constraints, and fulfils its intended use when placed in its intended environment.

1.3 Applicability

Refer to the table below for the applicability of this document.

Legal Entities	All RSBU Legal Entities
Line of Business (LoB)	Signalling and Rail Solutions (SRS)
Project/Bid Type/Criteria	Applies to all projects (R&D, Customer, Products & Solutions Roadmap, Service etc.)
Additional Criteria and Notes	-

1.4 Effective Date

This document is applicable as soon as it is published.

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2 TERMS

2.1 Acronyms and Abbreviations

CBS	Cost Breakdown Structure
DPMP	Development Project Management Plan
EMP	Engineering Management Plan
FM	Functional Manager
FMECA	Failure Modes, Effects and Criticality Analysis
FRACAS	Failure Reporting, Analysis, and Corrective Action System
GBMS	Global Business Management System
GPO	Global Process Owner
HSE	Health, Safety and Environment
ICD	Interface Control Document
IRT	Incident Response Team
JV	Joint Venture
LCC	Life Cycle Costing
LoB	Line of Business
PE	Project Engineer
PIRT	Product Incident Response Team
PJC	Project Controller
PJP	Project Planner
PM	Project Manager
PRM	Project RAMS Manager
RACI	Responsible, Accountable, Consulted and Informed
RAM	Reliability, Availability and Maintainability
RAMS	Reliability, Availability, Maintainability and Safety
REB/CEB	Regional Emergency Board/Central Emergency Board
RSBU	Rail Systems Business Unit
R&D	Research and Development
SRAC	Safety Related Application Condition

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SRS	Signalling and Rail Solutions (LoB)
S&M	Service and Maintenance
V&V	Verification and Validation
WBS	Work Breakdown Structure
WP	Work Package
WPL	Work Package Leader
WPR	Work PProduct

2.2 Definitions

WPL SIG	WPL of Design and/or Component and/or Generic Application and/or Specific Application and/or Verification and/or Validation related activities.
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2.3 GBMS Reference Documents

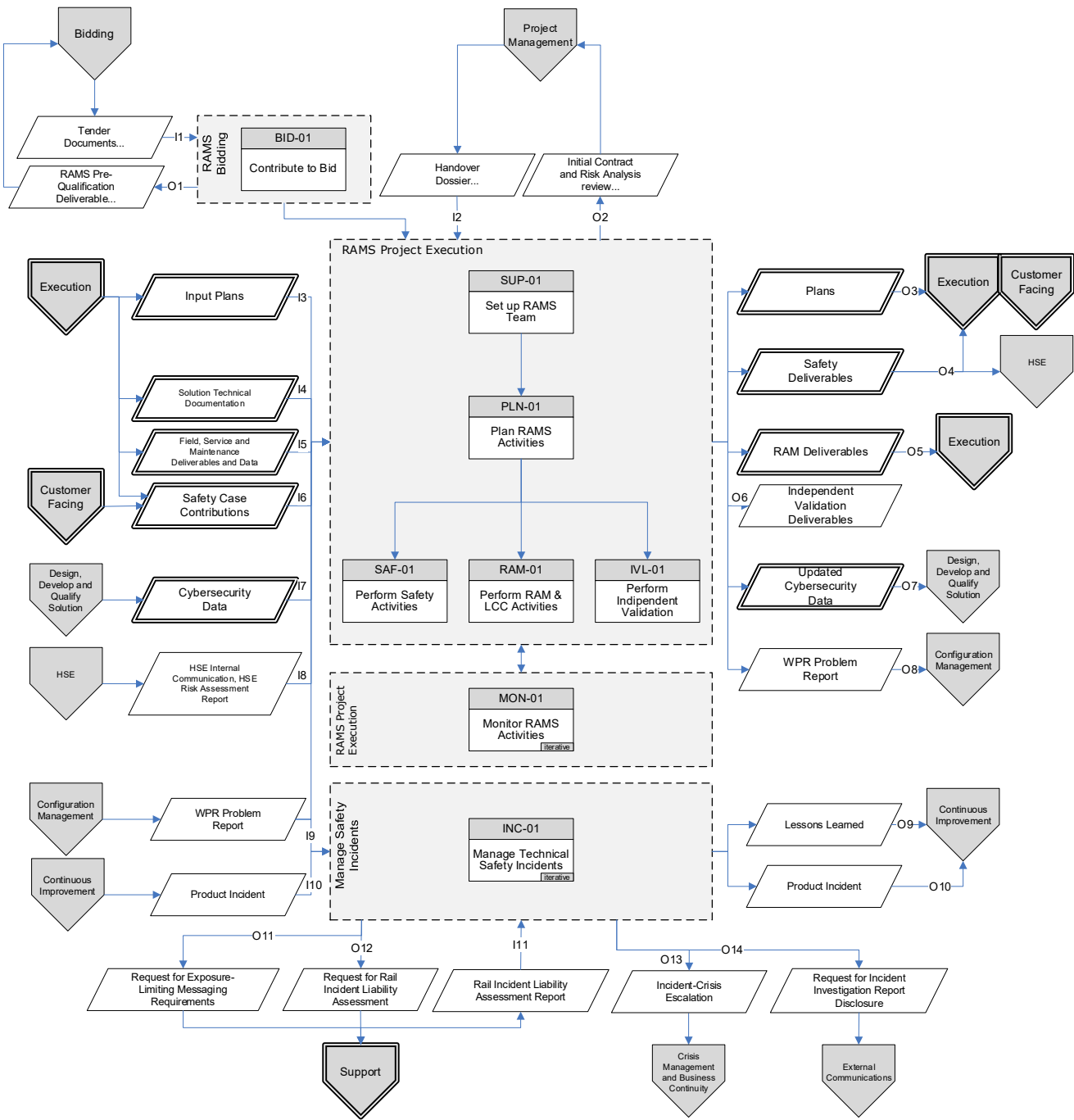
References apply to the latest approved revision.

G-PRC S0601*	Management of Project RAMS and Safety Related V&V Activities
G-PRC S0602*	Safety and Certification Activities
G-PRC S0603*	RAM & LCC Activities Management
G-PRC S06005	Manage Technical Safety Incidents

* under harmonisation

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3 PROCESS OVERVIEW



See §8 for input and output interface details.

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3.1 Entry and Exit Criteria

ENTRY CRITERIA	EXIT CRITERIA
RAMS Project Execution - Availability of formalized releases of Contractual documents. (Related inputs: I3, I4) - Availability of formalized release of Project Management documentation. (Related inputs: I3, I4) - Availability of formalized release of technical documentation. - Availability of Development Project Management Plan (DPMP). (I3) RAMS Bidding - Tender Documents, Bid Information, Deliverables and Templates, Consortium/JV Agreement, Draft Contract (if available). (I1) Manage Safety Incident - Product Incident. (I10) - WPR Problem Report. (I9)	RAMS Project Execution - Fulfilment of all RAM and Safety requirements at System/ Subsystem Level. - Gathering of all RAMS Reports. - Fulfilment of RAM and Safety targets during revenue service (if applicable). - Approval of contractual RAM and Safety Deliverables. - Handover to the Customer/SRS Asset Operation/SRS Service & Maintenance. RAMS Bidding - Technical Bid RAMS documentation. - RAMS bid estimate with related list of assumptions, risk & opportunities. - Conceptual bid solution approval according to RAMS strategy. - Bid RAMS Compliance Matrix. Manage Safety Incident - Product Incident. - WPR Problem Report

3.2 Inputs and Outputs

INPUTS	OUTPUTS			
I1 Tender Documents, Bid Information, Bid Deliverables List, Bidding Deliverables and Templates, Consortium/JV Agreement with other entities, Draft Contract (if available). I2 During Project Launch/Planning: Handover Dossier, Consolidated WBS and CBS, Budget, Second Level Schedule, Project Management Plan. Iteratively during Project Execution: Updated WBS/CBS/budget, Risk/Opportunity/Contingency, Second level schedule, Authorization of Change, Authorization of Delivery to Customer. I3 Input Plans.	O1 RAMS Pre-Qualification Deliverables, RAMS Bid Deliverables O2 <i>During Project Launch/Planning:</i> Initial Contract and Risk Analysis review, Budget up to Management Review/Management Approval, Functional Project Plans. <i>Iteratively during Project Execution:</i> Regular review of Project progress and deliverables, Functional Change Impact Estimation. O3 Plans. O4 Safety Deliverables. O5 RAM Deliverables. O6 Independent Validation Deliverables. O7 Updated Cybersecurity Data.			
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I4 Solution Technical Documentation.	O8 WPR Problem Report.
I5 Field, Service and Maintenance Deliverables and Data.	O9 ** Lessons Learned
I6 Safety Case Contributions.	O10 ** Product Incident
I7 Cybersecurity Data.	O11 ** Request for Exposure-Limiting Messaging Requirements.
I8 HSE Internal Communication, HSE Risk Assessment Reports.	O12 ** Request for Rail Incident Liability Assessment.
I9 WPR Problem Report	O13 ** Incident-Crisis Escalation.
I10 ** Product Incident	O14 ** Request for Incident Investigation Report Disclosure.
I11 ** Rail Incident Liability Assessment Report	

** to be confirmed

3.3 Work Environment Standard

* Tools have to be harmonised.

Product Document Management class (e.g. Teamcenter, ClearQuest).

Requirements Management class (e.g. IBM Doors or Jira for Hazard Log and SRAC Management).

Risk Management and System Reliability analysis (e.g. FaultTree++).

3.4 Verification Steps

Process correct implementation is verified by means of audits and, along the project developments, by gate reviews.

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4 ACTIVITIES

4.1 Contribute to Bid (BID-01)

(Reference docs: **G-PRC S0601***)

The PRM and the RAMS WPLs define the RAMS strategy and approve the conceptual Bid solution of the System/Subsystem accordingly. They produce the relevant Technical Bid RAMS and LCC (if applicable) documentation, RAMS bid estimate with related list of assumptions, risk & opportunities, to be submitted as an integral part of the overall Bid Proposal. Bid Qualification and Bid Submission respectively marks the beginning and the end of Bidding RAMS activities.

* under harmonisation.

4.2 Set up Project RAMS Team (SUP-01)

(Reference L2 docs: **G-PRC S0601***)

The Project RAMS Manager (PRM) is appointed. The PRM attends project kick-off meeting organized by PM and gathers contractual documents and tender documents handed over. Based on project Scope of Work, Work Packages and Project RAMS organization are defined. Project RAMS team starts working by analysing contract/project requirements.

* under harmonisation.

4.3 Plan RAMS Activities (PLN-01)

(Reference L2 docs: **G-PRC S0601***)

WPLs responsible for Independent Validation, RAM and Safety activities define the Independent Validation, RAM and Safety Plans. The PRM coordinates with the Independent Validation, RAM and Safety WPLs to detail the project schedule and the WBS, along with the project execution risk analysis. RAMS WPLs, based on bid assumption and contractual scope of work, confirm the operative budget and forecast workload related to their own WP, keeping the PRM informed.

* under harmonisation.

4.4 Perform Safety Activities (SAF-01)

(Reference L2 docs: **G-PRC S0601***, **G-PRC S0602***)

The PRM coordinates the execution of Safety activities, to achieve the expected results. The Safety WPLs are responsible to develop all Safety documents related to their owned WPs, based on project needs.

* under harmonisation.

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4.5 Perform RAM & LCC Activities (RAM-01)

(Reference L2 docs: **G-PRC S0601***, **G-PRC S0603***)

The PRM coordinates the execution of RAM activities, to achieve the expected results. The RAMs WPLs are responsible to develop all RAM documents related to their owned WPs, based on project needs.

* under harmonisation.

4.6 Perform Independent Validation (IVL-01)

(Reference L2 docs: **G-PRC S0601***)

The PRM coordinates the execution of Independent Validation activities, to achieve the expected results. The Independent Validation WPLs are responsible to develop all Independent Validation deliverables related to their owned WPs, based on project needs.

* under harmonisation.

4.7 Monitor RAMS Activities (MON-01)

(Reference L2 docs: **G-PRC S0601***)

The PRM or a dedicated Project Assurance Manager periodically (e.g. in support of approving safety lifecycle phase gates) monitors the completeness and evolution of the project RAMS execution and related V&V activities. This monitoring will be done via audit or inspection of phase completion artefacts with support from the project RAMS, Independent Validation, and SIG WPLs. Support and inputs from the System Validator, Project ISA and/or External Providers, if any, will also be relied upon to inform decisions related to phase completion approvals.

* under harmonisation.

4.8 Manage Technical Safety Incidents (INC-01)

(Reference L2 docs: **G-PRC S06005**)

The SRS RAMS assess and manage functional safety-related problems, ensuring the screening, classification and governing the timely evaluation, formal investigation and implementation of safety-related actions leading to the resolution of Technical Safety Incidents. The SRS RAMS shall capture lessons learned and establish preventive and improvement actions to minimize the likelihood and recurrence of safety-related incidents in products and projects. Lessons Learned are shared via the Continuous Improvement process for follow-up and implementation.

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5 TAILORING GUIDELINES

None.

6 MEASURES

Refer to **G-KPI S06001** that defines the KPIs associated with the process.

7 R.A.C.I.

Activity ID & Name	R = Responsible	A = Accountable	C = Consulted	I = Informed
BID-01	Tender RAMS WPLs, Tender PRM	Tender PRM	RAMS FMs, Tender PE	Bid Manager
SUP-01	PRM	RAMS FMs	PM, RAMS FMs	
PLN-01	RAMS and Independent Validation WPLs, System Validator	PRM	PJC, PJP, WPLs SIG	PM
SAF-01	Safety WPLs	PRM	System Validator, WPLs SIG	
RAM-01	RAM WPLs	PRM	System Validator, WPLs SIG, Maintenance Manager, SRS S&M WPL, SRS Asset Operation WPL	PM
IVL-01	Independent Validation WPLs	PRM	System Validator, WPLs SIG, Maintenance Manager, S&M WPL	PM
MON-01	RAMS and Independent Validation WPLs	PRM	System Validator, WPLs SIG	PM
INC-01	PIRT RAMS Referent, SIRT, RAMS WPL, PRM	Head of SRS RAMS	PIRT members, SRS RAMS Head of Business Segment/Region, Legal and Compliance unit, External Communications unit, Internal Communications unit, SRS Product QA unit, REB/CEB	SRS RAMS GPO, SRS RAMS Head of Region, PIRT members

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8 INPUT AND OUTPUT DETAILS

8.1 Input Details

ID	INPUT/INTERFACE		PROCESS CLUSTER			
	GENERIC	SPECIFIC	STEERING	SUPPORT	EXECUTION	CUSTOMER FACING
I1	-	- Tender Documents - Bid Information - Bid Deliverables List - Bidding Deliverables and Templates - Consortium/JV Agreement with other entities - Draft Contract (if available)	-	-	-	• Bidding
I2	-	During Project Launch/Planning: - Handover Dossier - Consolidated WBS and CBS - Budget - Second Level Schedule - Project Management Plan. Iteratively during Project Execution: - Updated WBS/CBS/budget, - Risk/Opportunity/Contingency, - Second level schedule - Authorization of Change - Authorization of Delivery to Customer.	-	-	-	• Project Management
I3	Input Plans	Requirements Management Plan, Interface Management Plan	-	-	• Design, Develop and Qualify Solution.	-
		Engineering Management Plan (EMP)	-	-	• Design, Develop and Qualify Solution	-
		Development Project Management Plan (DPMP)	-	-	• Design, Develop and Qualify Solution	-
I4	Solution Technical Documentation	Product Technical Documentation (including HW/SW Architecture and Requirements)	-	-	• Design, Develop and Qualify Solution	-
		System/Subsystem Architecture Definition and Requirements Specification, ICDs	-	-	• Design, Develop and Qualify Solution	-
I5	Field, Service and Maintenance Deliverables and Data	Failure Report			• SRS Service and Maintenance	
		Failure Report for FRACAS			• SRS Commissioning	

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ID	INPUT/INTERFACE		PROCESS CLUSTER			
	GENERIC	SPECIFIC	STEERING	SUPPORT	EXECUTION	CUSTOMER FACING
		Maintenance Resources, Line Replaceable Unit/Spare/Tool costs			• SRS Service and Maintenance	
		Repairing Data			• SRS Manufacturing	
I6	Safety Case Contributions	Verification Report, Validation Report	-	-	• Design, Develop and Qualify Solution	-
		Safety Case Part 2 (Quality Management Report)	-	-	-	• Project Quality Assurance
		Type Test Report	-	-	• Design, Develop and Qualify Solution	-
I7	Cybersecurity Data	Safety & Security Log	-	-	• Design, Develop and Qualify Solution	-
I8	-	HSE Internal Communication, HSE Risk Assessment Report	-	• HSE	-	-
I9		**WPR Problem Report (impacting Safety/for FRACAS)	-	-	• Configuration Management	-
I10		** Product Incident	-	• Continuous Improvement	-	-
I11	-	** Rail Incident Liability Assessment Report		• Legal and Compliance		

** to be confirmed

8.2 Output Details

ID	OUTPUT/INTERFACE		PROCESS CLUSTER			
	GENERIC	SPECIFIC	STEERING	SUPPORT	EXECUTION	CUSTOMER FACING
O1	-	RAMS Pre-Qualification Deliverables, RAMS Bid Deliverables	-		-	• Bidding
O2	-	During Project Launch/Planning: - Initial Contract and Risk Analysis review - Budget up to Management Review/Management Approval - Functional Project Plans.	-		-	• Project Management

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ID	OUTPUT/INTERFACE		PROCESS CLUSTER			
	GENERIC	SPECIFIC	STEERING	SUPPORT	EXECUTION	CUSTOMER FACING
		Iteratively during Project Execution: - Regular review of Project progress and deliverables - Functional Change Impact Estimation				
O3	Plans	RAM Plan	-	-	-	• Project Management
		**Reliability Growth Plan	-	-	• SRS Commissioning • SRS Service and Maintenance	
		**FRACAS and RAM Demonstration Plan	-	-	• SRS Commissioning • SRS Service and Maintenance	-
		Safety Plan	-	-	• Design, Develop and Qualify Solution	• Project Management
		Independent Validation Plan	-	-	-	-
		External Safety/V&V Audit yearly programme	-	-	-	• Project Quality Assurance
O4	Safety Deliverables	Hazard Log	-	-	• Design, Develop and Qualify Solution • SRS Asset Operation • SRS Service and Maintenance	-
		**SRAC register	-	-	• SRS Asset Operation • SRS Service and Maintenance	-
		Safety Case	-	• HSE	-	-
		Technical Safety Note	-	• HSE	• SRS Commissioning	-
O5	RAM Deliverables	FRACAS procedures	-	-	• SRS Asset Operation	-
		FRACAS specification/template	-	-	• SRS Service and Maintenance	-
		RAM & LCC Data	-	-	• SRS Industrialisation	-
		** Preliminary FMECA	-	-	• Design, Develop and Qualify Solution	-
		RAM and performance report	-	-	• SRS Service and Maintenance	-
		RAM/LCC Analysis	-	-	• SRS Service and Maintenance	-

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ID	OUTPUT/INTERFACE		PROCESS CLUSTER			
	GENERIC	SPECIFIC	STEERING	SUPPORT	EXECUTION	CUSTOMER FACING
O6	Independent Validation Deliverables	Independent Validation Report	-	-	-	-
O7	Updated Cybersecurity Data	Safety & Security Log Updated	-	-	Design, Develop and Qualify Solution	-
O8	-	WPR Problem Report	-	-	Configuration Management	-
O9	-	**Lessons Learned	-	Continuous Improvement	-	-
O10	-	** Product Incident		Continuous Improvement	-	-
O11	-	** Request for Exposure-Limiting Messaging Requirements		- External Communications - Human Resources	-	-
O12	-	** Request for Rail Incident Liability Assessment		Legal and Compliance	-	-
O13	-	** Incident-Crisis Escalation	Crisis Management and Business Continuity	-	-	-
O14	-	** Request for Incident Investigation Report Disclosure	-	External Communications	-	-

** to be confirmed

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